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Senior Capstone - Data Science
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Video Game: Project Egress

Project Proposal:

Our plan is to make a video game titled Project Egress. Project Egress will be a 2-D game that is similar to Portal (not with the portal concept, just the level system) where you pass through multiple short rooms or levels. The game will be almost like a system of trial chambers in some sort of grey facility to bring out the colors of paint the player character will be using. The player character will have different 'paints' and weapons or tools to use said paints in order to reach the goal set in each level. Each weapon will provide a different method to splatter the paint and each paint will have a different effect.

More specifically, our game will be a puzzle-solving game, a platformer, a shooter, and a sidescrolling game. There should be roughly 3 to 4 different paint colors that will have different effects such as green being sticky, red being slippery or fast, and orange being bouncy. For the weapons, each weapon will fire a different size projectile and different rates (balancing) and we might even include entirely different tools all together if possible. We will be using 'WASD' for movement and left click for shooting. Our planned keybinds are entirely subject to change.

We will make our project with Unity which uses C#. In terms of roles, Ethan will be making the art and models for the game using a pixel art website which will later be ported to unity and will be contributing to the code to the game by making edits or helping out where we have any shortcomings in the coding process. Liam will be doing the majority of the coding for the game. Madeline will be helping with both the code and art. Our milestones will be: finishing a 2D sandbox environment, finishing pixel art, creating basic objectives, applying the code and sprites, completing the game.

Our first main goal will be to get the paint system working and have the movement for the player feel good. After that, our main goal will be to make a level for each paint color. Overall, our main goal is to at least have a functioning game regardless of whether we polish it to the extent of having fully functional levels and a clean user interface. It is highly possible that we

will achieve a sandbox, it is just a matter of how much time it will take to achieve everything. Our main priority is to be able to get all of our basic functions and art in the game so it is still enjoyable to play. Ideally, since the levels and art will be relatively simple in art and color, the main focus will be just getting the characters movements and actions to work first and foremost.

Languages/Software:

Languages:

- C#

Software:

- Unity
- Piskel
- Visual Studio Code